SAEON Sprint Backlog

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

Planning and evidence basis

Member	Primary responsibility
Redah Gamieldien	Project Leader and hardware integration lead
Qaasim Isaacs	Backend Developer and Tester
Angelo Adams	Frontend Developer and Designer
Ethan Williams	Analyst Researcher and Writer
Rameez Karriem	Agile Lead and Presenter

Sprint overview and proposed effort

Week / PB	Focus	Proposed h
1 / PB01	Project Initiation & System Architecture	10
2 / PB02	Core Firmware & Sensor Interfacing	14
3 / PB03	HiveMQ Cloud Broker Setup	11
4 / PB04	Cellular Communication (SIM7000G)	14
5 / PB05	SAEON Dashboard Base Application	15
6 / PB06	Power Supply & Solar Integration	12
7 / PB07	Advanced UI & Health Diagnostics	14
8 / PB08	Hardware Finalization & PCB Soldering	17
9 / PB09	Pre-Deployment Auditing & Calibration	13
10 / PB10	48-Hour Field Stress Test	12
11 / PB11	Presentation Formatting & Content	12
12 / PB12	Rehearsal & Submission Prep	8
13 / PB13	Project Buffer & Final Review	4

Sprint 1 Project Initiation & System Architecture

PB01 | 28 July 2026 to 31 July 2026 | Tuesday to Friday

Sprint goal

Agree the project scope, verify components, set up development tools and document the architecture.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
1.01	Define Project Scope & Board Setup	Redah	2	Completed
1.02	Component Verification & Sourcing	Redah	3	Completed
1.03	Development Environment Configuration	Qaasim	2	Completed
1.04	Architecture Diagram & Data Flow	Ethan	3	Completed

Acceptance criteria

- Scope, roles, component choices and architecture are documented.
- Trello task tracking and GitHub evidence responsibilities are clear.
- The configuration distinguishes DHT11 testing, the final SHT31 sensor and Firebase storage.

Sprint 2 Core Firmware & Sensor Interfacing

PB02 | 4 August 2026 to 7 August 2026 | Tuesday to Friday

Sprint goal

Test DHT11 readings and rain-gauge tip counting, prepare low-power firmware and document the SHT31 requirements for the final product.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
2.01	ESP32 Base Setup & Serial Output	Redah	3	Completed
2.02	DHT11 Testing and SHT31 Final Sensor Requirements	Redah	4	Completed
2.03	TR-525M Rain Gauge Logic	Redah	4	Completed
2.04	Firmware Formatting & Low Power Mode	Redah	3	Completed

Acceptance criteria

- Controlled rain-gauge tips are counted once using a verified conversion factor.
- DHT11 test readings and invalid-sensor states are identifiable; records identify the sensor used.
- Low-power behaviour has bench-test records. Final SHT31 validation is tracked separately under PB17.

Sprint 3 HiveMQ Cloud Broker Setup

PB03 | 11 August 2026 to 14 August 2026 | Tuesday to Friday

Sprint goal

Configure HiveMQ access, test MQTT delivery and validate the backend ingestion route into Firebase.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
3.01	Broker Provisioning	Qaasim	2	Completed
3.02	Topic Structuring & Access Control	Qaasim	2	Completed
3.03	Network Connection & Publish Testing	Qaasim	3	Completed
3.04	Firebase Storage Integration	Qaasim	4	Completed

Acceptance criteria

- HiveMQ access uses documented publish and subscribe permissions.
- Sample readings pass through the MQTT route securely.
- Firebase stores station-linked, timestamped telemetry.

Sprint 4 Cellular Communication (SIM7000G)

PB04 | 18 August 2026 to 21 August 2026 | Tuesday to Friday

Sprint goal

Send real readings through SIM7000G cellular MQTT and check connection stability.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
4.01	SIM Module Wiring & Power	Redah	3	Completed
4.02	Network Bearer Configuration	Qaasim	3	Completed
4.03	Cellular MQTT to Firebase Integration	Redah	4	Completed
4.04	Stability & Error Handling	Qaasim	4	Completed

Acceptance criteria

- Real messages reach the dashboard over cellular with Wi-Fi disabled.
- Network registration and modem configuration are recorded.
- Outage recovery and duplicate handling are checked.

Sprint 5 SAEON Dashboard Base Application

PB05 | 25 August 2026 to 28 August 2026 | Tuesday to Friday

Sprint goal

Build the dashboard, live displays, historical filters and the interface for planned cloud commands.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
5.01	App Skeleton & Routing	Angelo	4	Completed
5.02	Real-Time Visualization	Angelo	5	Completed
5.03	Historical Data & Filtering	Angelo	4	Completed
5.04	Cloud Command Prep	Qaasim	2	Completed

Acceptance criteria

- Current readings show units and latest update times.
- Historical filters and data views work against stored records.
- Cloud-command preparation is documented without claiming remote-control acceptance.

Sprint 6 Power Supply & Solar Integration

PB06 | 1 September 2026 to 4 September 2026 | Tuesday to Friday

Sprint goal

Integrate battery and solar charging and verify the alternative power arrangement.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
6.01	Battery & Charge Controller Setup	Redah	4	Completed
6.02	Solar Panel Integration	Redah	3	Completed

Task	Expected work	Lead	Proposed h	Status
6.03	Mains Power Alternative Testing	Qaasim	2	Completed
6.04	System Power Flow Audit	Qaasim	3	Completed

Acceptance criteria

- Battery and charging components are integrated and checked.
- Solar and alternative power test results are recorded.
- The power-flow audit records limitations.

Sprint 7 Advanced UI & Health Diagnostics

PB07 | 8 September 2026 to 11 September 2026 | Tuesday to Friday

Sprint goal

Provide station diagnostics, CSV export, responsive views and an end-to-end pipeline check.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
7.01	Develop Station Health Diagnostics UI Component	Angelo	4	Completed
7.02	Implement CSV Export Functionality	Qaasim	3	Completed
7.03	Mobile & Tablet Responsiveness	Angelo	3	Completed
7.04	Full End-to-End Data Pipeline Ping Test	Qaasim	4	Completed

Acceptance criteria

- Station health and stale-data states are visible.
- CSV export matches the selected data.
- Responsive views and the complete data path are checked.

Sprint 8 Hardware Finalization & PCB Soldering

PB08 | 15 September 2026 to 18 September 2026 | Tuesday to Friday

Sprint goal

Transfer the prototype to its permanent circuit and mount it in the prepared enclosure.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
8.01	Circuit Transfer & Layout	Redah	4	Completed
8.02	PCB Soldering	Redah	6	Completed
8.03	Enclosure Preparation	Redah	3	Completed
8.04	Component Mounting & Sealing	Redah	4	Completed

Acceptance criteria

- The permanent circuit matches the reviewed wiring.
- Soldering and mounting are checked.
- The enclosure is prepared without treating assembly as field acceptance.

Sprint 9 Pre-Deployment Auditing & Calibration

PB09 | 22 September 2026 to 25 September 2026 | Tuesday to Friday

Sprint goal

Audit power and charging, then proceed with installation and baseline field calibration.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
9.01	Power Drain Audit	Qaasim	3	Completed
9.02	Solar Charging Verification	Redah	2	Completed
9.03	Full System Deployment	Redah	4	In Progress
9.04	Baseline Calibration	Redah	4	In Progress

Acceptance criteria

- Power drain and solar charging checks are Completed.
- Installation and baseline field calibration remain In Progress.
- Final site permission, protection and field records are required before acceptance.

Sprint 10 48-Hour Field Stress Test

PB10 | 29 September 2026 to 2 October 2026 | Tuesday to Friday

Sprint goal

Complete and audit the 48-hour field stress test; this field-validation work remains In Progress.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
10.01	Stress Test Initiation	Qaasim	2	In Progress
10.02	Active Monitoring	Qaasim	3	In Progress
10.03	Manual Field Measurements	Ethan	3	In Progress
10.04	Data Auditing & Conclusion	Ethan	4	In Progress

Acceptance criteria

- The test duration, uptime and delivery gaps are recorded when available.
- Reference measurements and timestamps can be compared.
- Field conclusions are not marked Completed before the test and audit finish.

Sprint 11 Presentation Formatting & Content

PB11 | 6 October 2026 to 9 October 2026 | Tuesday to Friday

Sprint goal

Complete presentation content, diagrams, dashboard evidence and technical documentation.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
11.01	Presentation Outlining	Ethan	2	Completed
11.02	Technical Diagrams & Architecture Slides	Ethan	3	Completed
11.03	Dashboard & Software Slides	Ethan	3	Completed
11.04	Technical Documentation	Ethan	4	Completed

Acceptance criteria

- Slides describe the actual architecture and current results.
- The setup guide and technical documentation are complete for the prototype.
- Open field-validation limitations are explicit.

Sprint 12 Rehearsal & Submission Prep

PB12 | 13 October 2026 to 16 October 2026 | Tuesday to Friday

Sprint goal

Finalise the slides, rehearse the demonstration and compile the prototype submission materials.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
12.01	Slide Deck Finalization	Ethan	2	Completed
12.02	Group Rehearsal 1	Rameez	2	Completed
12.03	Timed Rehearsal & Feedback	Rameez	2	Completed
12.04	Final Compilation	Rameez	2	Completed

Acceptance criteria

- Speaking parts and rehearsals are prepared.
- A backup demonstration and submission package are compiled.
- Preparation does not imply a submission receipt or final field approval.

Sprint 13 Project Buffer & Final Review

PB13 | 20 October 2026 to 23 October 2026 | Tuesday to Friday

Sprint goal

Review the prototype documentation package and record outstanding deployment work separately.

Tasks and proposed effort

Task	Expected work	Lead	Proposed h	Status
13.01	Buffer & Prototype Documentation Sign-off	Redah	4	Completed

Acceptance criteria

- The prototype documentation and presentation package are reviewed.
- The deployment and field-validation actions remain open.
- Prototype sign-off is distinguished from field-deployment acceptance.